Never Knew How Much That Muddy Water Meant To Me: Satellite Derived Suspended Sediment Monitoring of the Chattahoochee Watershed

Ilan Valencius[1] (valencig@bc.edu), Noah P. Snyder[1], Evan Dethier[2], Mark Behn[1], Christopher Sheehan[1], Travis Dahl[3]

[1] Boston College, Department of Earth and Environmental Sciences, Chestnut Hill, MA, [2] Dartmouth College, [3] US Army Corps Engineering Research and Development Center

Abstract

The dynamics of river suspended sediment, derived from soil erosion, is critical for understanding floodplain and coastal wetland evolution, as well as reservoir sedimentation. Although the U.S. Geological Survey (USGS) has conducted >10^5 suspended sediment concentration (SSC) sampling campaigns, data availability is often sparse or altogether lacking for large river transects. Landsat derived SSC measurements have proven accurate enough to supplement USGS datasets, allowing unprecedented spatial analysis of SSC trends throughout large river systems[1].Here, we build on this approach by applying it to higher spatial and temporal resolution datasets. We have derived suspended sediment concentrations from the Sentinel-2 sensor through a cluster and regression approach. To increase the number of training samples, we constructed SSC-discharge rating curves for all in-situ USGS stations. This has constrained the uncertainty of Sentinel-2 derived SSC to less than a factor of two, which has proven adequate for large rivers. In combination with the Landsat record, this allows for a multi-decadal analysis of sediment transport dynamics across multi-dammed systems. We apply these methods to the Chattahoochee River in Georgia and Alabama, USA. Using observations from 1984 to 2022, we find pronounced decreases in SSC downstream of dams along the river, with downstream reaches never regaining the same values as upstream observations. We also see a decreasing trend in SSC across the whole river reach, which could be indicative of changing land-use practices.

Sentinel-2 Regression

125 USGS stations were used to create the calibration set[2,3]. To minimize bank effects these stations are located on channels of ≥ 90m width as determined via the HydroSHEDS database[1,4,5].To harmonize with Sentinel-2 data, a sample had to be taken within 4 days of a Sentinel-2 image[1]. To increase the number of samples, a SSC vs. discharge rating curve was constructed for every station with ≥10 in-situ samples. Stations with an $R^2 \geq 0.8$ were kept. Discharge is monitored daily so this increased the number of training samples from 9,926 to 25,732. The rating curve derived samples come from 118 unique stations over the period 12-10-2018 to 07-21-2022.

After the rating-curve derived samples were harmonized with Sentinel-2 imagery downloaded from Google Earth Engine[6], every band ratio was added as well as the square and square root of every raw band. This is to capture any potential non-linear reflectance response as SSC varies. Using KMeans clustering (n=6) each individual station was assigned a cluster. 10-fold cross-validation lasso regression was then used for each cluster group to derive the coefficients for the regression equation[1]:

$$\log_{10}[SSC(mg/L)] = Intercept_i + \sum_{n=0}^{n_{max}} Coef_n * Variable_n$$

By performing L1 shrinkage, fewer, more descriptive, coefficients are selected with the rest being set to 0. To account for the fact that SSC spans orders of magnitude, the relative error across all clusters was used to determine the accuracy of the model. Relative error is calculated as

$$\text{relative error} = 10^{M\left(\log_{10}\left|\frac{SSC_{predicted}}{SSC_{in\ situ_i}}\right|\right)} - 1$$

where M denotes the median and $SSC_{predicted}$ and $SSC_{in\ situ_i}$ are drawn from the holdout set not used for calibration[1,7]. 20% of samples in every cluster were taken as the holdout set.

The Chattahoochee Watershed

The US Army Corps Engineering Research and Development Center Coastal Hydraulics Laboratory has been investigating reservoir sedimentation in the Chattahoochee-Apalachicola Basin basin. Landsat and Sentinel-2 imagery offers a more comprehensive picture of sediment transport than traditional methods[1]. To derive flux measurements, a daily discharge vs. drainage area rating curve was constructed using the USGS gauging stations present on the main stem of the Chattahoochee – defined here as the extent ranging from Buford Dam to the intersection of the Apalachicola and Brothers river, a 711 km length. This allows for a continuous estimate of discharge along the main stem transect and a continuous estimate of sediment flux along the Chattahoochee River. Using this, the trapping efficiencies of the Walter F. George and West Point dam have been estimated as 0.45 and 0.49-0.83.

Figure 4: Suspended sediment concentrations along the Chattahoochee transect from 1985-present. Landsat derived SSC measurements are derived from Dethier et al. (2020) while the USGS data comes from gauging stations on the main stem of the transect.

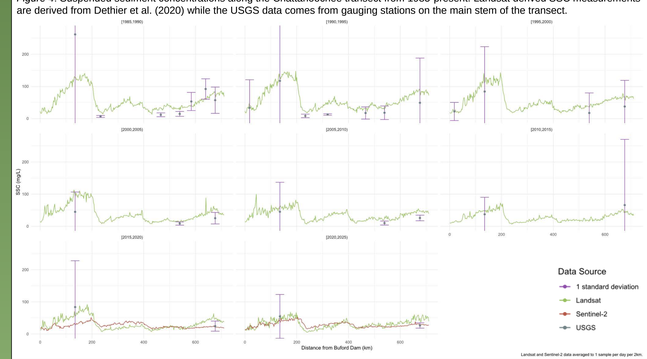
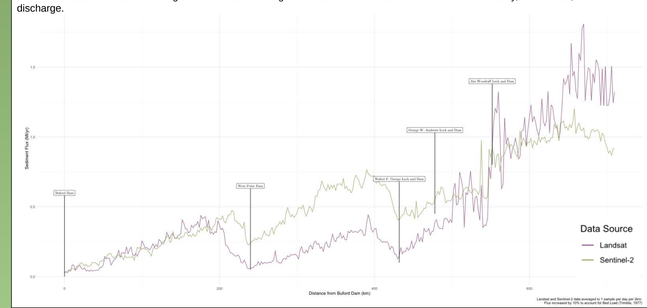


Figure 5: Sediment flux along the Chattahoochee transect from 2015-present. Flux is determined by coupling satellite derived suspended sediment measurements with rating curve derived discharge from USGS in-situ stations. This allows for a daily, continuous, estimate of discharge.



Anthropogenic Impacts on Sediment

The net sediment flux of a watershed is crucial for wetlands and coastal ecosystems. We observe three distinct time periods describing the sediment history of the southeastern U.S.[10]:

Pre-Colonial → (pre-1700s) There is little large scale farming practices or logging. Anthropogenic contributions to the sediment budget of the watershed are minimal.

Pre-Dam → (1700s – early 1900s) Intensive land clearing and agriculture occurred along with the construction of many small dams on tributaries.

Post-Dam → (mid-1900s – present) All 5 major US Army Corps of Engineers dams were completed from 1954-1974.

Although sediment has been trapped by dams, we observe that there is still an increase in net sediment flux compared to the pre-anthropogenic time period. The amount of sediment being impounded in reservoirs is crucial for understanding what effect land use practices have had on the sediment budget.

Figure 7: Sentinel-2 image of West Point Lake and dam on March 6, 2019. The image offers a visual illustration of how reservoirs trap sediment. The change in discharge above and below the dam is minimal so SSC offers a reasonable proxy of sediment flux through the reservoir. Mean SSC is determined via the Sentinel-2 calibration.

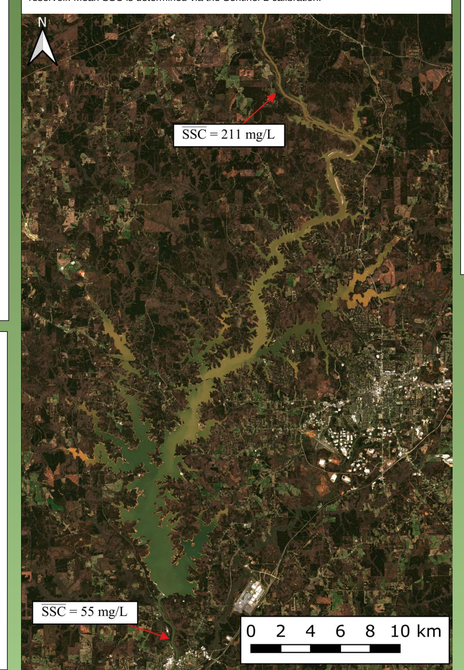


Figure 9: Major dams in the Chattahoochee Watershed[9].

Figure 10: How anthropogenic factors have affected sediment dynamics in the southern Appalachian Piedmont[10].

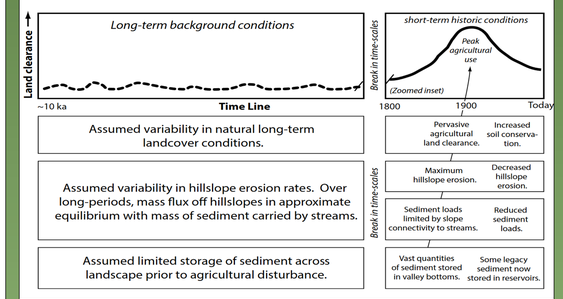


Figure 11: Sediment flux from the Chattahoochee watershed. Pre-anthropogenic estimates are from Be^{10} data[10,11]. USGS data is from the most downstream gauging station, site number 02359170, Apalachicola River near Sumatra FL. For Landsat and Sentinel-2 data, the mouth of the river is determined to be the last 10km of the transect. Pre-dam estimates are derived from in-situ measurements taken over the course of 1909[12]. Post dam estimates agree with the various sources compiled by Milliman (2013) within error[13].

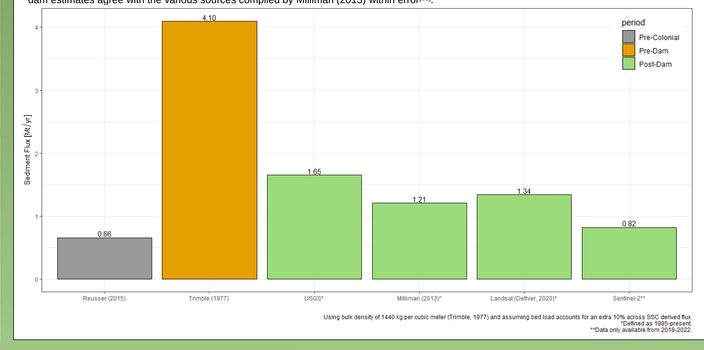


Figure 1: Median "true color" appearance of various SSC levels (via USGS rating curve data) across 6 Sentinel-2 river groupings

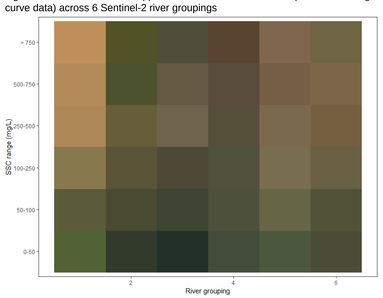


Figure 2: Lasso regression performance on a 20% holdout set, red line indicates perfect prediction. Note the model overpredicts at low SSC ranges due to bottom effects.

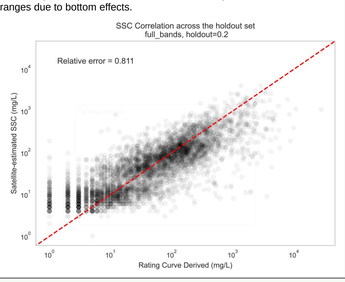


Figure 3: USGS sites used for Sentinel-2 clustering and regression[2,3,4,5,6]. Note: one station in Alaska was removed for clarity.

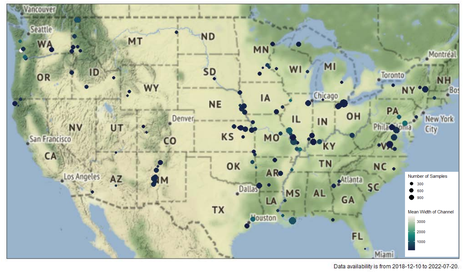


Figure 6: Comparison of data density at the mouth of the Chattahoochee, 1985-2022.

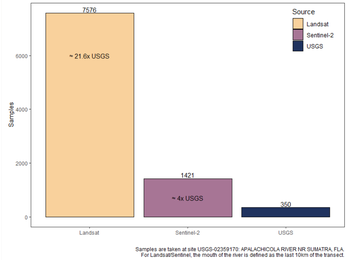
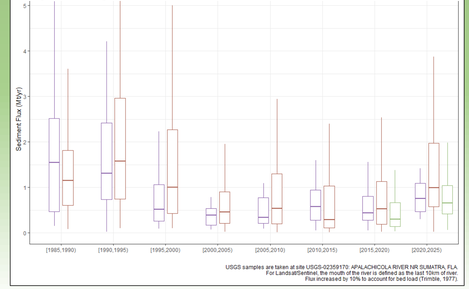


Figure 8: Average sediment flux at the mouth of the Chattahoochee, 1985-2022.



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